

GRAND FORKS INTL, ND, USA

WMO: 727570

Lat: 47.943N

Lon: 97.184W

Elev: 842

StdP: 14.25

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14916

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.6	-16.8	-27.5	1.2	-20.9	-22.9	1.6	-16.3	32.5	4.2	28.9	8.0	7.4	300	0.681

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.2	88.9	71.9	85.9	69.9	83.3	68.3	75.1	84.7	72.9	82.5	70.7	79.8	13.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.0	121.9	81.6	69.6	112.3	79.0	67.3	103.6	76.3	39.2	84.9	37.0	82.4	35.1	79.5	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.4	24.6	21.4	-28.2	94.7	7.0	2.3	-33.2	96.3	-37.3	97.6	-41.2	98.9	-46.3	100.6
			-27.6	78.9	6.0	2.0	-31.9	80.3	-35.4	81.5	-38.7	82.6	-43.1	84.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.6	6.9	10.7	24.7	41.3	54.5	65.3	69.7	67.7	58.9	44.3	27.6	13.3
	DBStd	24.58	14.61	13.84	13.58	10.16	8.86	6.86	5.52	5.85	8.45	9.21	11.51	13.06
	HDD50	5622	1337	1100	786	288	55	1	0	0	19	227	673	1136
	HDD65	9345	1802	1520	1249	710	340	78	21	39	219	645	1121	1601
	CDD50	2179	0	0	2	29	195	459	610	548	286	49	1	0
	CDD65	427	0	0	0	0	15	86	166	122	36	2	0	0
	CDH74	4659	0	0	0	15	260	933	1684	1295	437	35	0	0
	CDH80	1415	0	0	0	2	74	276	523	405	129	6	0	0
Wind	WSAvg	10.4	11.0	11.1	11.1	11.4	11.3	9.8	8.5	8.7	9.8	10.8	10.7	10.8
Precipitation	PrecAvg	19.4	0.6	0.5	0.9	1.3	2.3	3.2	3.0	2.6	2.0	1.4	0.8	0.6
	PrecMax	25.2	1.8	1.7	3.1	3.6	5.7	7.9	7.5	6.2	5.6	5.8	3.9	2.0
	PrecMin	12.6	0.1	0.0	0.0	0.0	0.2	0.7	0.5	0.4	0.3	0.1	0.0	0.1
	PrecStd	3.5	0.5	0.4	0.6	1.0	1.4	1.9	1.9	1.3	1.4	1.3	0.8	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.2	44.3	63.0	75.4	87.6	92.3	91.6	92.2	89.0	79.0	60.5	41.9
		MCWB	36.4	38.8	52.7	55.9	65.9	72.5	75.2	74.3	70.6	60.6	50.2	36.4
	2%	DB	36.0	38.4	53.3	69.5	81.1	86.2	88.3	88.0	83.9	70.5	53.2	36.9
		MCWB	33.3	35.1	46.6	53.3	61.5	68.7	73.5	71.0	67.0	57.6	45.6	33.8
	5%	DB	32.0	34.9	46.6	63.8	75.6	82.9	85.5	84.4	79.3	64.0	47.9	33.8
		MCWB	30.1	32.9	41.8	50.7	58.6	66.7	71.5	69.5	64.5	54.2	42.1	31.7
	10%	DB	27.5	31.2	41.1	58.8	71.4	79.6	82.6	81.3	74.3	59.2	42.7	30.7
		MCWB	25.9	29.4	37.9	48.1	56.8	65.4	69.4	67.4	62.2	51.1	38.4	28.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.5	39.8	55.0	58.7	68.3	76.6	78.4	77.1	73.4	64.1	51.4	37.4
		MCDB	39.7	43.9	60.4	71.3	81.5	86.3	87.8	88.2	85.4	74.3	57.1	41.2
	2%	WB	33.4	35.5	47.3	54.3	64.7	72.5	75.8	73.7	69.4	59.4	46.5	34.0
		MCDB	35.6	37.6	52.9	66.4	75.3	82.1	84.8	84.1	79.1	68.0	52.5	36.5
	5%	WB	30.0	32.9	41.8	51.5	61.6	69.4	73.5	71.4	66.6	55.2	42.3	31.7
		MCDB	32.0	34.9	46.2	62.2	72.0	79.0	82.6	80.9	75.1	63.3	47.3	33.7
	10%	WB	26.0	29.2	37.8	48.9	59.0	67.2	71.4	69.3	64.0	51.2	38.4	29.1
		MCDB	27.4	31.2	41.0	58.3	68.4	75.7	80.2	78.3	72.5	58.1	42.5	30.7
Mean Daily Temperature Range	5% DB	MDBR	16.9	18.2	17.0	20.7	23.3	21.4	22.2	23.5	23.0	19.2	16.3	15.5
		MCDBR	18.4	17.8	22.7	30.9	30.2	26.4	25.3	27.2	29.5	28.5	23.9	18.4
		MCWBR	16.5	15.0	16.7	17.9	15.2	12.7	13.1	13.1	14.9	16.9	16.7	15.6
	5% WB	MCDBR	18.3	17.0	22.0	28.3	24.7	22.2	22.9	23.9	24.9	26.4	22.7	17.7
		MCWBR	16.5	14.4	16.5	17.1	14.1	12.5	13.3	13.1	14.5	16.5	16.3	15.3
Clear-Sky Solar Irradiance	taub	0.283	0.301	0.301	0.353	0.369	0.383	0.383	0.384	0.349	0.306	0.277	0.265	
	taud	2.286	2.258	2.440	2.356	2.373	2.362	2.365	2.363	2.454	2.545	2.517	2.414	
	Ebn at Noon	249	271	293	285	284	279	277	272	272	271	258	247	
	Edh at Noon	23	30	30	36	37	38	37	36	30	23	19	18	
All-Sky Solar Radiation	RadAvg	501	883	1241	1546	1741	1901	2012	1679	1232	763	451	345	
	RadStd	53	79	129	141	112	141	87	115	87	110	53	40	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page